

STAT 312: HW Assignment 2 - Unit 03

Instructor: Dr. Md Mutasim Billah

Submission Guidelines

Please follow these guidelines carefully when completing and submitting your homework. Clear and well-organized work helps ensure fair and accurate grading.

- There are **6 problems** in total.
- Submit all homework assignments through **Canvas**.
- If you prefer to work with **pen and pencil**:
 - Download and/or print the homework assignment.
 - Scan your completed work and combine all pages into a **single PDF file**.
 - Upload the PDF to the corresponding homework assignment on Canvas.
- If you prefer to use a **writing device** or type your solutions, you may complete the homework electronically and submit your final work directly as a **PDF**.
- You may also need to submit an **R file** if the homework includes a programming task.
- Your work must be **clear, organized, and legible**. Please follow these presentation guidelines:
 - Clearly circle or otherwise mark your final answers so they can be easily located on the page.
 - Show all necessary work for **nontrivial calculations**. Answers without sufficient supporting work may receive little or no credit.
- Name your submitted PDF using the following format:
`firstname_lastname_HW#.pdf`
For example: `john_smith_HW#.pdf`

Problem 2.1 - Conditional Probability and Tree Diagram

Thomas Muse Guitars, a music shop in Cleveland, sells both acoustic and electric guitars. Some have a price less than \$500 and some do not. If you randomly select a guitar in the music shop, the probability it is an acoustic guitar is 0.40. If you randomly select an acoustic guitar, the probability its price is less than \$500 is 0.64. If you randomly select an electric guitar, the probability its price is less than \$500 is 0.34.

First, define the following events: A denotes an acoustic guitar, E denotes an electric guitar, L denotes a guitar priced less than \$500, and $\bar{L}$ denotes a guitar priced \$500 or more.

Using these probabilities and event definitions, **draw a tree diagram** and then answer the following questions.

(a) If you select a guitar at random, what is the overall probability that it costs \$500 or more?

(b) If you randomly select a guitar among those that cost less than \$500, what is the probability that it is an electric guitar?

Problem 2.2 - Venn Diagram and Conditional Probability

If a book at Kelvin Smith Library is selected at random, the probability it is fiction is $P(F) = 0.61$. The probability it is a hardcover book is $P(H) = 0.74$. The probability it is neither fiction nor hardcover (in other words, the book is a non-fiction paperback) is 0.05.

Let F = fiction, $\overline{F}$ = non-fiction, H = hardcover, and $\overline{H}$ = paperback. Draw a **Venn diagram** using the given information and then answer the following questions.

- (a) What is the probability that a randomly selected book is fiction and a paperback?
- (b) What is the probability that a randomly selected book is non-fiction or a paperback?
- (c) Given that a randomly selected book is a hardcover, what is the probability that it is fiction?

Problem 2.3 - Mutually Exclusive and Independent Events

Let A and B be two events. Suppose, $P(A) = 0.70$, and $P(B) = 0.20$.

(a) If A and B are **mutually exclusive**, find $P(A \cup B)$.

(b) If A and B are **independent**, find $P(A \cup B)$.

Problem 2.4 - Craps

A gambling game named Craps requires a player to roll two fair dice and add the two numbers that appear. Suppose one die is red and the other is blue. There are 36 equally likely outcomes in the sample space:

(red = 1, blue = 1)

(red = 2, blue = 1)

(red = 3, blue = 1)

$\vdots$

(red = 5, blue = 6)

(red = 6, blue = 6)

The table below has 36 cells corresponding to these 36 outcomes. The sum corresponding to each outcome appears in the appropriate cell.

		<i>red die</i>					
		1	2	3	4	5	6
<i>blue die</i>	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

Find each of the following probabilities. **Leave your final answer as a simplified fraction.**

(a) $P(\text{sum} = 5) =$

(b) $P(\text{sum} = 6 \cup \text{blue} = \text{red}) =$

(c) $P(\text{blue} = 6 \cap \text{sum} = 5) =$

(d) $P(\text{blue} = \text{red} \mid \text{sum} = 4) =$

(e) $P(\text{red} = 4 \mid \text{red} \leq \text{blue}) =$

(f) $P(\text{sum is an even number} \mid \text{blue} < 4) =$

For each pair of events below, identify the events as **independent**, **mutually exclusive**, or **neither**. Justify your answer **mathematically** (as we did in the class).

(g) sum is an odd number and $\text{blue} \leq 3$

(h) $\text{sum} = 5$ and $\text{blue} = \text{red}$

(i) $\text{red} = 2$ and $\text{red} \leq \text{blue}$

(j) $\text{sum} = 7$ and $\text{blue} = 5$

(k) $\text{red} = 3$ and $\text{sum} = 11$

Problem 2.5 - Probability with Replacement

Bill has 54 candies in a bag: 29 are red, 14 are blue, and 11 are white. He will draw three candies out of the bag one at a time at random **with replacement**.

- (a) Find the probability that Bill will draw 1 red candy and 2 white candies in this exact order: red-white-white.

- (b) What is the probability that Bill will draw **at least one blue candy**?

Problem 2.6 - Tree Diagram and Bayes' Rule

Thousands of college students are attending a political conference and are about to enjoy a dinner banquet. The probability that a randomly selected student is a Democrat is 0.70 and the probability that a randomly selected student is a Republican is 0.30. Before the banquet, students were asked what they wanted for dinner. Their options were a vegetarian dish, beef, or chicken. Among the Democrat students, the proportion who requested the vegetarian dish was 0.20, the proportion who requested beef was 0.30, and the proportion who requested chicken was 0.50. Among the Republican students, the proportion who requested the vegetarian dish was 0.10, the proportion who requested beef was 0.50, and the proportion who requested chicken was 0.40.

Let D = Democrat, R = Republican, V = vegetarian dish, B = beef, and C = chicken. Draw a **tree diagram** using the information above. Then use the tree diagram to answer the following questions.

- (a) If a student is selected at random, what is the probability they requested the vegetarian dish for dinner?

- (b) Given that a randomly selected student requested beef for dinner, what is the probability they are a Republican?
- (c) Given that a randomly selected student did not request the vegetarian dish, what is the probability they are a Democrat?

GOOD LUCK!